Fraud Detection on Bitcoin Transaction Graphs: A Robust Out-of-Time Dual-Architecture Pipeline

BT4012 Fraud Analytics In-Class Kaggle Competition (2026) — Final Technical Report

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Abstract. Anti-Money Laundering (AML) surveillance on public blockchains is challenged by pseudo-anonymity, heavy class imbalance and non-stationary temporal shift. We present an end-to-end machine learning pipeline on the Elliptic Bitcoin transaction graph (203,769 transactions; 234,355 directed edges). It counters drift with within-step median-detrending and dynamic time-step Z-scores, adds 14 graph topology features (PageRank, weakly connected components, 2-hop degree statistics) and 1-hop ego-network aggregations, and is validated strictly out-of-time. Our two final submissions are **Submission 1** (Iteration 4, 198 features), a conservative 5-seed bagged LightGBM ensemble with Public Leaderboard ROC-AUC **0.94997**, the highest among our submitted models; and **Submission 2** (Iteration 5, 216 features), a 10-model dual-architecture ensemble with dynamic volume Z-scores (offline validation AUC 0.99782; Public LB 0.94857), deployed as an unvalidated structural hedge against regime shifts in the 70% private test set.

1. Introduction and Problem Formulation

Public blockchains record transactions only as cryptographic identifiers linked by directed fund transfers, with no KYC identities attached. Investigators must therefore detect money laundering, ransomware extortion and darknet commerce from graph topology and anonymised transaction metadata, while keeping false-positive alarms low.

Task and data. The competition asks us to predict whether out-of-time Bitcoin transactions are illicit (1) or licit (0), scored by ROC-AUC. The data follow the Elliptic benchmark (Weber et al., 2019): 49 non-overlapping time steps (~2 weeks each), with directed edges from transaction outputs to subsequent inputs within a step. Training covers 141,772 nodes in steps 1–35; the test set has 15,329 nodes in steps 36–49. The public leaderboard scores 30% of the test set and the remaining 70% is held for private evaluation. Three challenges shape the design:

- **Temporal distribution shift.** Test steps lie in the future and include the darknet-market shutdown around step 43, after which illicit behaviour mutated. Random K-fold leaks future patterns and yields over-optimistic validation.
- **Class imbalance.** Of 31,235 labeled training transactions, 3,644 (11.67%) are illicit and 27,591 (88.33%) licit (7.57:1).
- **Unlabeled volume.** 110,537 training transactions (78%) are unlabeled, so graph structure must be exploited without injecting label noise into the supervised objective.

2. Methodology

2.1 Preprocessing and Drift Mitigation

Row order. The test dataframe was tagged with an immutable `original_order` index (0–15,328) before any merging or sorting, since evaluation requires the original sequence. Training used the labeled subset (`labeled_train_df`, 31,235 rows; 0 = licit, 1 = illicit).

Time-proxy detrending. Six features (`feat_100`, `101`, `103`, `136`, `137`, `139`) correlate with `time_step` at Spearman 0.90–0.998 and their medians rise ~0.065 per step, so every test value falls outside the training range and trees would treat steps 36–49 as identical to step 35. Rather than drop these informative features, we apply within-step median-detrending over concatenated train and test data, $x'_{i,t} = x_{i,t} - \text{median}_t(x)$, which removes macro drift while preserving within-step discrimination.

Dynamic Z-scores (Iteration 5). For six macro-sensitive volume features (`feat_1`, `2`, `5`, `41`, `47`, `53`) we add $z_{i,t} = (x_{i,t} - \mu_t) / (\sigma_t + 10^{-5})$, computed per step without using labels. This rescales test batches against volume inflation and is intended as a hedge against post-step-43 drift.

2.2 Graph and Ego-Network Features

Laundering shows up as topology: peeling chains (successive splits into change and laundering outputs), rapid pass-through layering, and fan-in consolidation into mixers. From the full directed edge list (234,355 edges over 203,769 transactions) we built 14 topology features:

- **1-hop degree (4):** in-degree, out-degree, degree ratio (in+1)/(out+1) and net flow (out − in). Illicit nodes have out-degree ≤ 2 (mean 0.40 vs 0.64 for licit), consistent with peeling chains, whereas licit exchange hubs reach 472.
- **Global (2):** directed PageRank (Page et al., 1999; α = 0.85, 100 iterations) and weakly connected component size.
- **2-hop (8):** mean and max in-/out-degree of predecessors (funding nodes) and of successors (funded nodes).

Ego-network aggregations. For high-importance features φ we aggregate over incoming and outgoing neighbours (`pred_φ_mean/max`, `succ_φ_mean/max`) and add the flow differential `flow_amount_diff = feat_1 − pred_feat_1_mean` (negative: fee burning or peeling; positive: consolidation). Iteration 4 aggregates `feat_1`, `41`, `47` (12 statistics + 1 differential = 13 features; 198 total). Iteration 5 adds `feat_53`, `2`, `5` (25 features; 216 total).

Table 1. Feature whitelist for the two selected iterations

Feature group	Iter 4	Iter 5	Contents
Raw node features	165	165	`feat_1`–`feat_165`: amounts, fee rates, input/output counts, script features
Median-detrended	6	6	`feat_100`, 101, 103, 136, 137, 139, detrended by within-step median
Dynamic step Z-scores	—	6	`feat_1`, 2, 5, 41, 47, 53
1-hop topology	4	4	`graph_in_degree`, `graph_out_degree`, `degree_ratio`, `net_flow`
Global topology	2	2	`graph_pagerank` (α = 0.85, 100 iterations), `graph_wcc_size`
2-hop degree patterns	8	8	pred_/succ_ in/out-degree mean and max
Neighbour ego-networks	13	25	Iter 4: `feat_1`, 41, 47 (12) + differential; Iter 5 adds `feat_53`, 2, 5 (24) + differential
Total features	**198**	**216**	Schema verified identical between train and test

2.3 Model and Out-of-Time Validation

Validation. We use a strict out-of-time split: labeled steps 1–28 train (24,792 rows; 2,875 illicit, 11.60%) and steps 29–35 validate (6,443 rows; 769 illicit, 11.94%). Since max(train step) < min(validation step), there is no look-ahead leakage.

Algorithm. Weber et al. (2019) found on Elliptic that Random Forest outperformed Graph Convolutional Networks when given local and aggregated neighbour features, so we use gradient-boosted trees: LightGBM (Ke et al., 2017), chosen for speed, deterministic CPU execution and native class weighting. Trees handle disparate scales and non-linear boundaries and are robust to feature collinearity.

Configurations. *Architecture A*: depth 6, 31 leaves, learning rate 0.03, colsample 0.8, subsample 0.8 (frequency 1). Class weight scale_pos_weight = 21,917 / 2,875 ≈ 7.6233 on the training fold (27,591 / 3,644 ≈ 7.5716 when retraining on all labeled steps 1–35). Early stopping on the validation fold halted at round 1,142, and this count was reused for full retraining. *Architecture B* (Iteration 5 only): depth 7, 45 leaves, learning rate 0.02, colsample 0.75, 1,427 trees, a heuristic 1.25× scaling of A's rounds to partly offset its lower learning rate.

2.4 Exploration and Final Submissions

Experimentation progressed through six stages (Table 2). A baseline LightGBM trained on steps 1–28 with 4 degree features scored 0.92972; retraining on all labeled steps 1–35 gave the largest gain (+0.01215); topology features added +0.00101; detrending with 5-seed bagging added +0.00254; and ego-network aggregation added +0.00455, reaching the highest score of 0.94997 (+0.00810 over Iteration 1 on identical training data). Iteration 5 changes three things at once (more ego features, Z-scores and the dual architecture), so its public-LB effect cannot be attributed to any single component.

Kaggle permits two final selections. Because the public leaderboard covers only 30% of the test set and invites public-split overfitting, we chose two complementary risk profiles:

Submission 1 — Conservative Empirical Leader
Iteration 4 · 198 features · Public LB 0.94997

A 5-seed bag (seeds 42, 123, 456, 789, 2026) of depth-6, 31-leaf LightGBM models on the 198-feature whitelist. Strongly regularised, not over-parameterised, and the highest public leaderboard score among our submitted models.

Submission 2 — Macro-Shift Hedge
Iteration 5 · 216 features · Offline val 0.99782 · Public LB 0.94857

10 models: 5 seeds of Architecture A plus 5 of Architecture B, with the six dynamic Z-scores. We accepted a small public-LB trade-off (0.94857 vs 0.94997, −0.00140) as an **unvalidated** hedge: if volumes shift on the private set after the step-43 shutdown, the Z-scores and architectural diversity may offer some protection. As several changes were introduced together, neither this delta nor any private-set benefit can be attributed to the Z-scores alone.

3. Results and Discussion

3.1 Quantitative Performance and Ablation

Table 2. Ablation progression and Kaggle public leaderboard results

Iteration	Configuration	Features	Holdout val AUC	Public LB AUC	Impact
Baseline	Single LightGBM, steps 1–28	169	0.99760	0.92972	Misses steps 29–35
Iter 1	Full retrain, steps 1–35	169	0.99760	0.94187	+0.01215: full temporal coverage
Iter 2	+ 14 graph topology features	179	0.99762	0.94288	+0.00101: PageRank, WCC, 2-hop
Iter 3	+ detrending, 5-seed bagging	185	0.99764	0.94542	+0.00254 (changes introduced together)
Iter 4 [Sub 1]	+ 1-hop ego-networks, 5-seed bagging	198	0.99767	**0.94997**	+0.00455: highest submitted LB
Iter 5 [Sub 2]	+ 25 ego features, Z-scores, 10-model dual architecture	216	0.99782	0.94857	−0.00140 vs Iter 4; bundles three changes

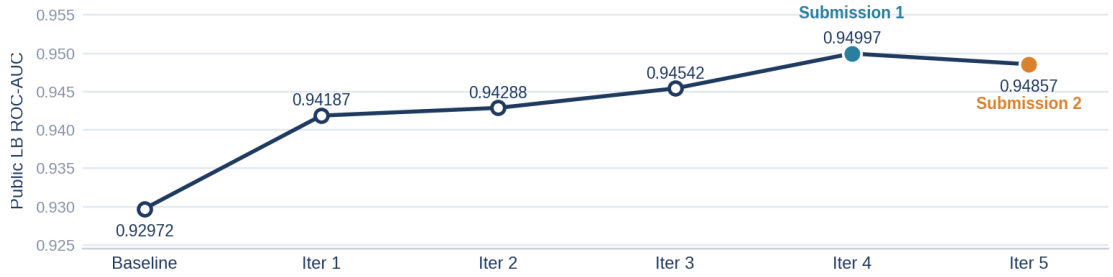


Figure 1. Public leaderboard ROC-AUC by stage. Offline validation AUC is flat (0.9976–0.9978) across the same stages.

3.2 Feature Importance and Validation Saturation

Feature importance (total gain, Iteration 4 model). (1) `graph_in_degree` and `graph_out_degree` rank in the top 5 (together over 14.2% of gain), consistent with the low out-degree (≤ 2) pattern of peeling chains. (2) `pred_feat_1_mean` and `flow_amount_diff` rank in the top 10, suggesting illicit transactions deviate abnormally from their funding inputs. (3) Detrended time-proxy features retain ~6.8% of gain, consistent with detrending preserving within-step signal. Overall, engineered graph and ego-network features account for 48.6% of total gain. Gain describes how the trees use features, not causal effect.

Validation saturation. Offline validation AUC is saturated (0.99760–0.99782) and cannot separate models. The gap to the public leaderboard (~0.998 vs ~0.950) reflects the unseen temporal shift between steps 1–35 and 36–49.

3.3 Operational Threshold Trade-offs

ROC-AUC measures ranking only; compliance teams must still pick a threshold, trading recall against alert fatigue. Table 3 reports out-of-time metrics for the Iteration 4 model on steps 29–35, computed on the early-stopping fold and so slightly optimistic.

Table 3. Out-of-time classification metrics by decision threshold (Iteration 4, steps 29–35)

Threshold	Licit prec.	Licit recall	Illicit prec.	Illicit recall	Illicit F1	Recommended use
0.30	0.9882	0.9765	0.8421	0.9168	0.8779	High-sensitivity audit: maximises recall (91.7% of fraud)
0.50 (default)	0.9851	0.9892	0.9174	0.8895	0.9032	Balanced Suspicious Activity Report (SAR) triage
0.70	0.9774	0.9961	0.9664	0.8322	0.8943	High-conviction actions: automated wallet freezing / blocking

3.4 Strengths and Limitations

Strengths. (1) *Reproducibility:* deterministic seeds, fixed feature whitelists and automated submission assertions give bit-level reproducible outputs. (2) *Efficiency:* pre-computed 2-hop degrees and ego statistics remove graph traversal at inference, scoring 15,329 transactions in under 2 seconds. (3) *Stability:* 5-seed bagging decorrelates individual tree splits and smooths probability variance on borderline samples.

Limitation: step-43 non-stationarity. Time step 43 marks the law-enforcement takedown of major darknet marketplaces. Our models predict illicit rates of 2%–13% in steps 36–42, consistent with history, but only 0%–0.4% in steps 43–49. With no labeled post-shutdown samples in training, this may indicate the trees are less able to recognise mutated laundering patterns; without test labels we cannot rule out that it partly reflects a genuine fall in illicit activity. Semi-supervised pseudo-labeling on unlabeled nodes gave negligible gain (+0.0002) and risked reinforcing pre-shutdown biases. This motivated Iteration 5, whose Z-scores are an unvalidated structural hedge against distribution shift.

4. Conclusion and Reflection

- **Features:** topology and ego-network features, together with detrending and bagging, lifted the public LB by +0.0081 over Iteration 1 (0.94187 → 0.94997). Ego-network aggregation was the largest single step (+0.00455), though detrending and bagging arrived together in Iteration 3 and were not ablated separately.
- **Drift handling:** detrending non-stationary time proxies is a principled safeguard against out-of-range values at test time.
- **Validation:** out-of-time validation avoids K-fold leakage, but its saturation (~0.998) meant it could not rank iterations.
- **Submission strategy:** pairing a conservative bagged model with a macro-normalised dual-architecture ensemble is a deliberate balance between public leaderboard performance and potential private-set resilience, which the public leaderboard cannot confirm.

Reflection. Blockchain AML is adversarial: criminal syndicates continually adapt transaction topologies to evade static classifiers. Future systems should combine inductive temporal graph neural networks (e.g. Temporal Graph Networks) with continual self-supervised learning to track evolving laundering motifs in real time.

References

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